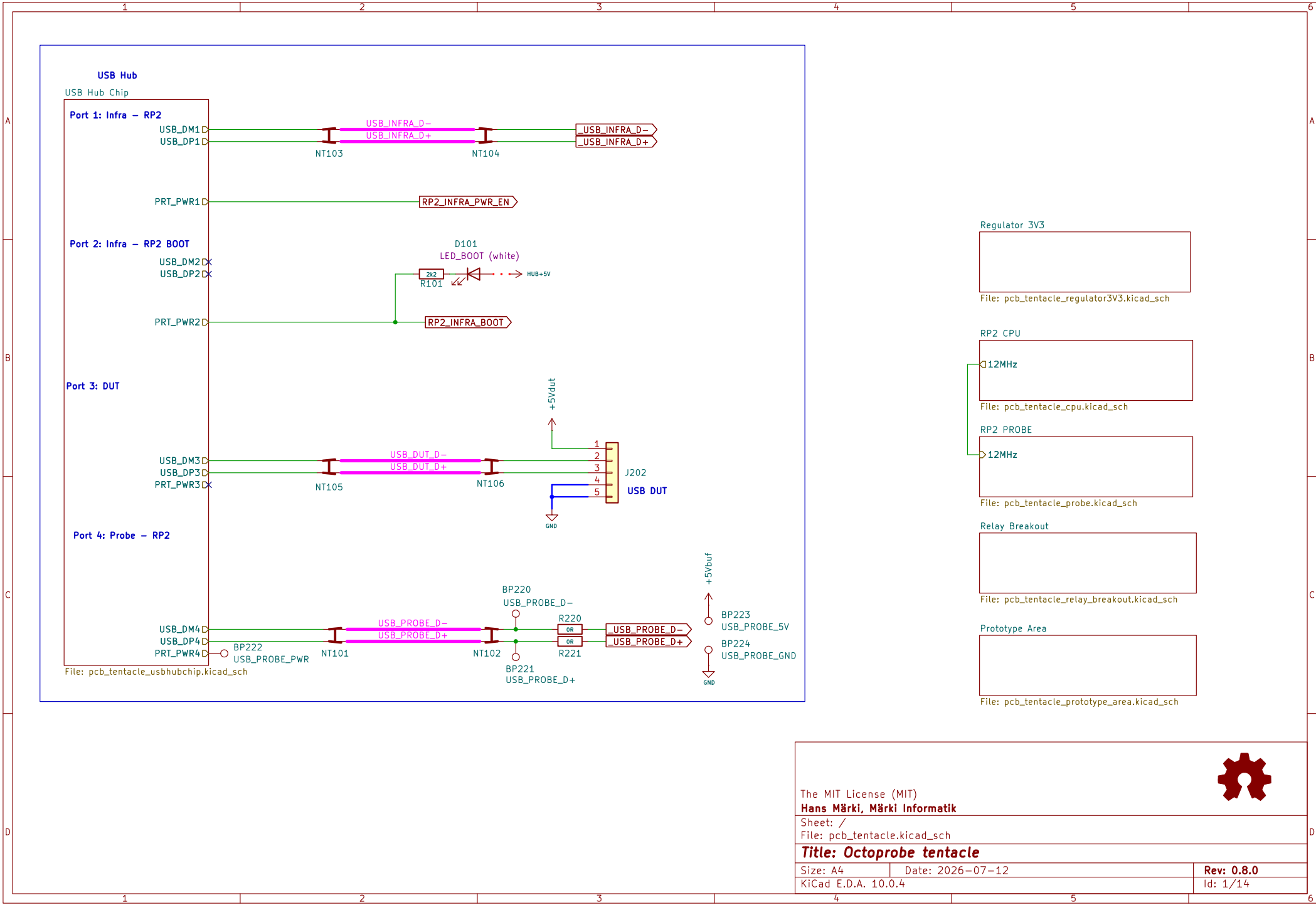
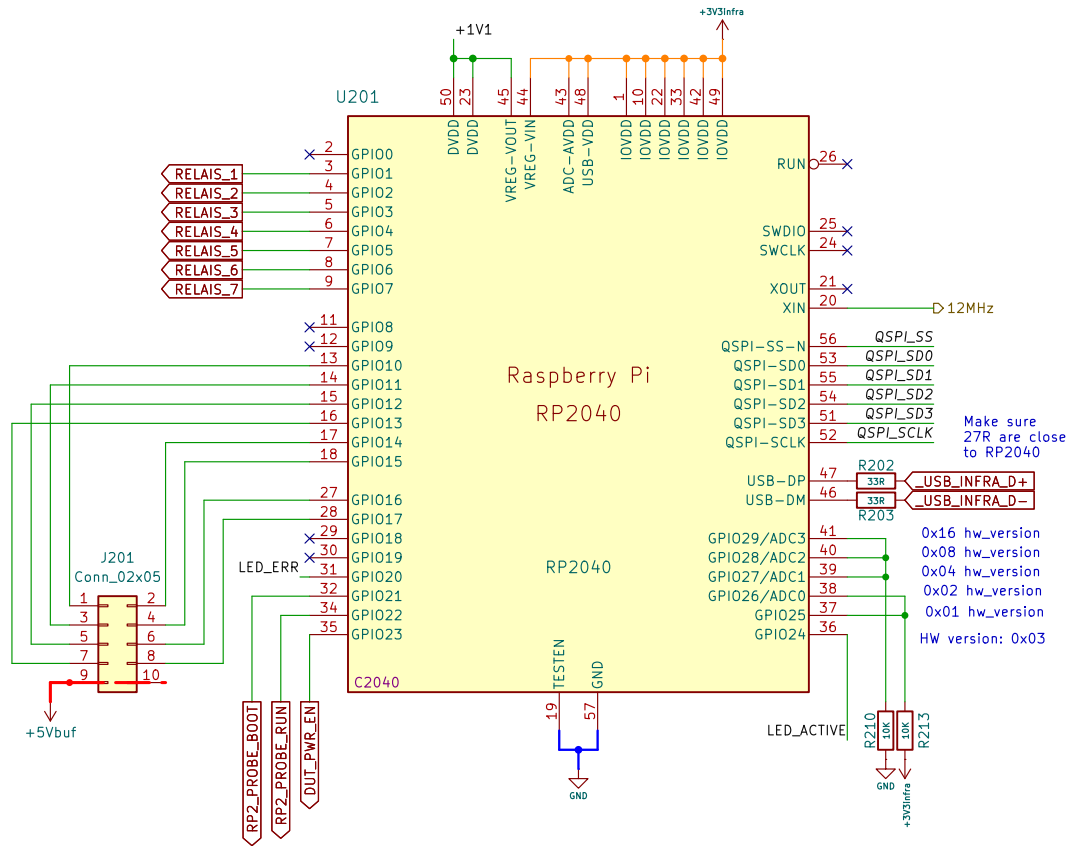
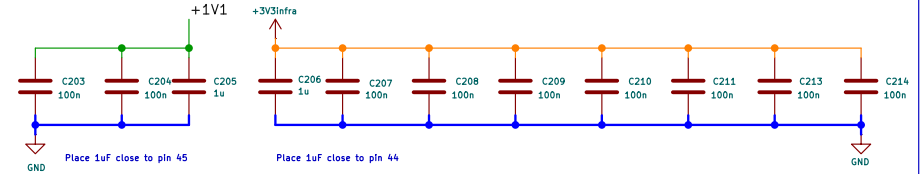




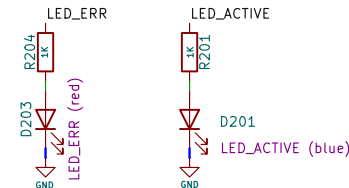
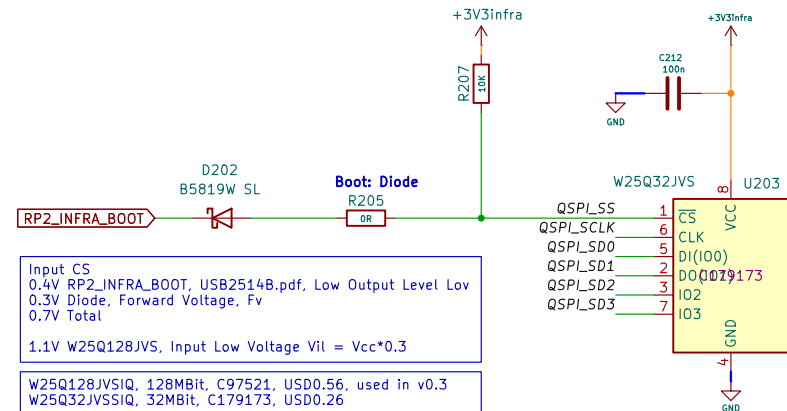
RP2040 Infra



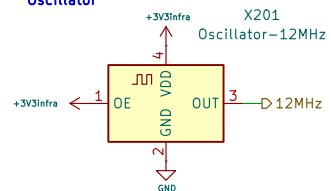
Bypass Capacitors



Flash Memory



Oscillator



JLC alternatives

Jlc	USD/Stock	Package	Comments
C2831468	0.37/1k	SMD2520-4P	1nd
C2901596	0.37/1k	SMD2520-4P	2nd
C437139	0.29/188	SMD2520-4P	2nd
C387449	0.49/3k	SMD2520-4P	2nd
C409301	0.55/2k	SMD2520-4P	2nd

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Sheet: /RP2 CPU/

File: pcb_tentacle_cpu.kicad_sch

Title: Octoprobe tentacle

Size: A4 Date: 2026-07-12

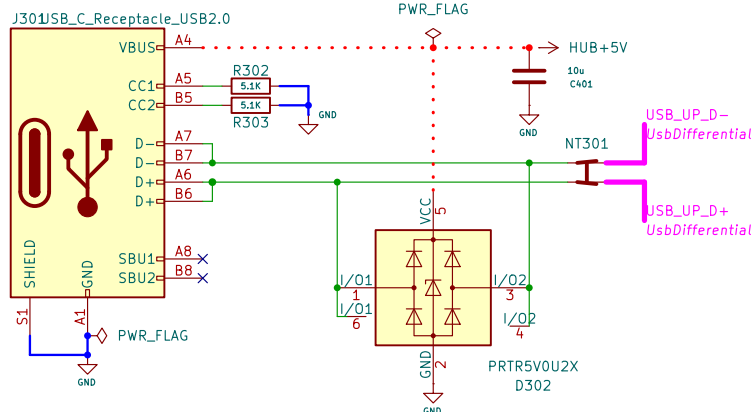
KiCad E.D.A. 10.0.4

Rev: 0.8.0

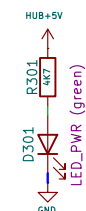
Id: 2/14

USB C Female

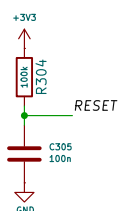
USB Type-C DATA



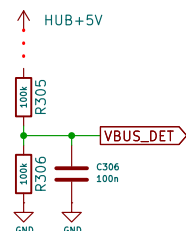
Power LED



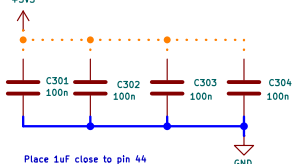
Boot Sequence



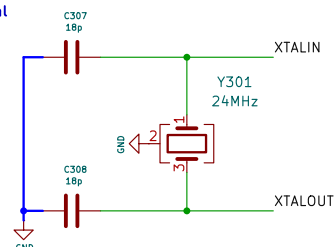
VBUS_DET



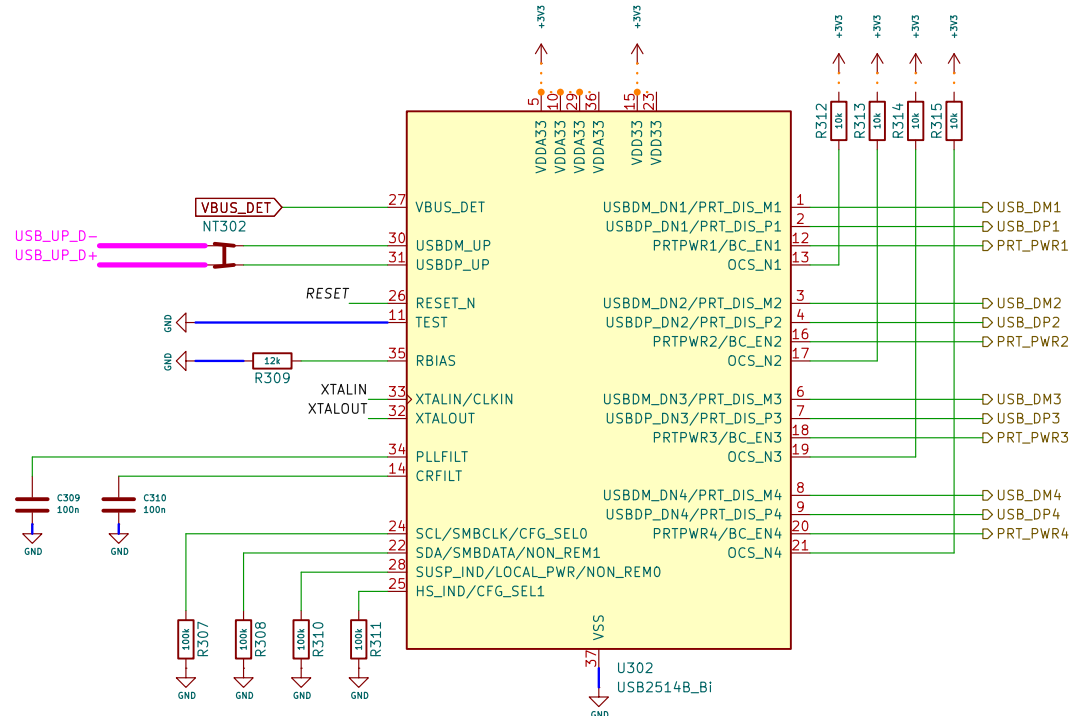
Bypass Capacitors



Crystal



Hub



Scope GND Pad



Microchip AN 15.17 "PCB Layout Guide for USB Hubs"

DP/DM:
27mil, 0.7mm, trace
5mil, 0.127mm, space
50mil, 1.27mm, trace side no ground

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Sheet: /USB Hub Chip/

File: pcb_tentacle_usbhubchip.kicad_sch

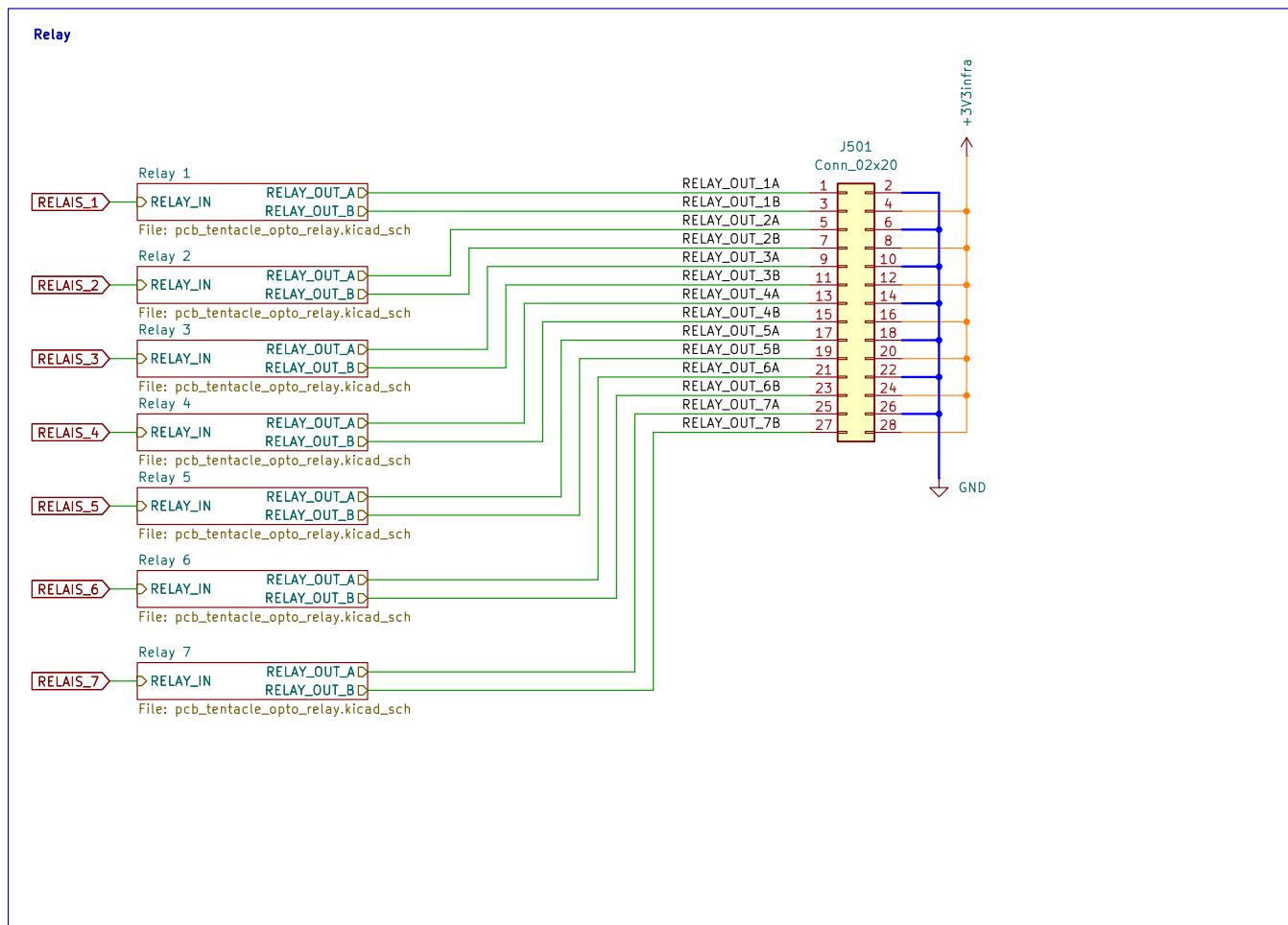
Title: Octoprobe tentacle

Size: A4 Date: 2026-07-12

KiCad E.D.A. 10.0.4

Rev: 0.8.0

Id: 3/14



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Sheet: /Relay Breakout/

File: pcb_tentacle_relay_breakout.kicad_sch

Title: Octoprobe tentacle

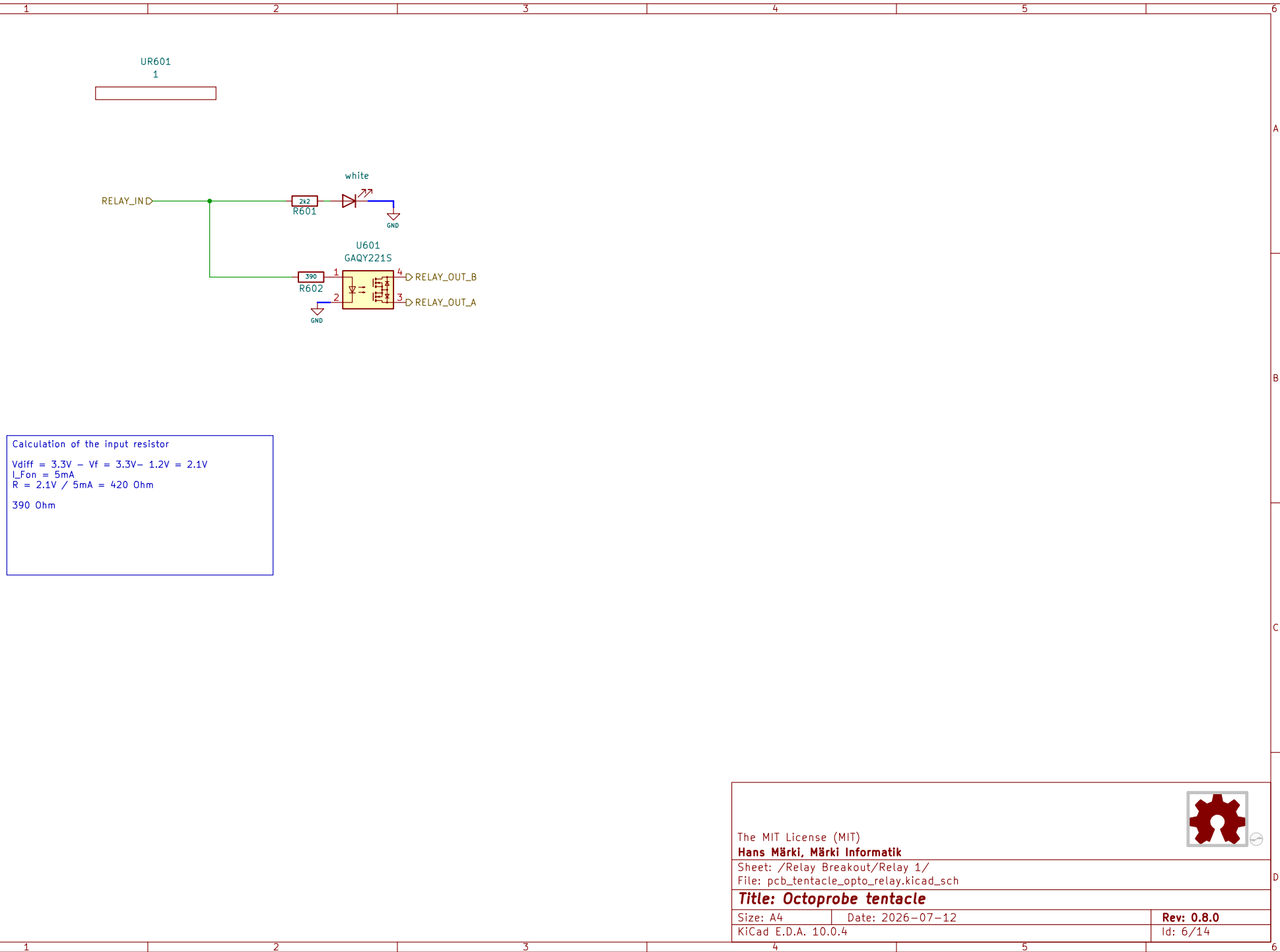
Size: A4

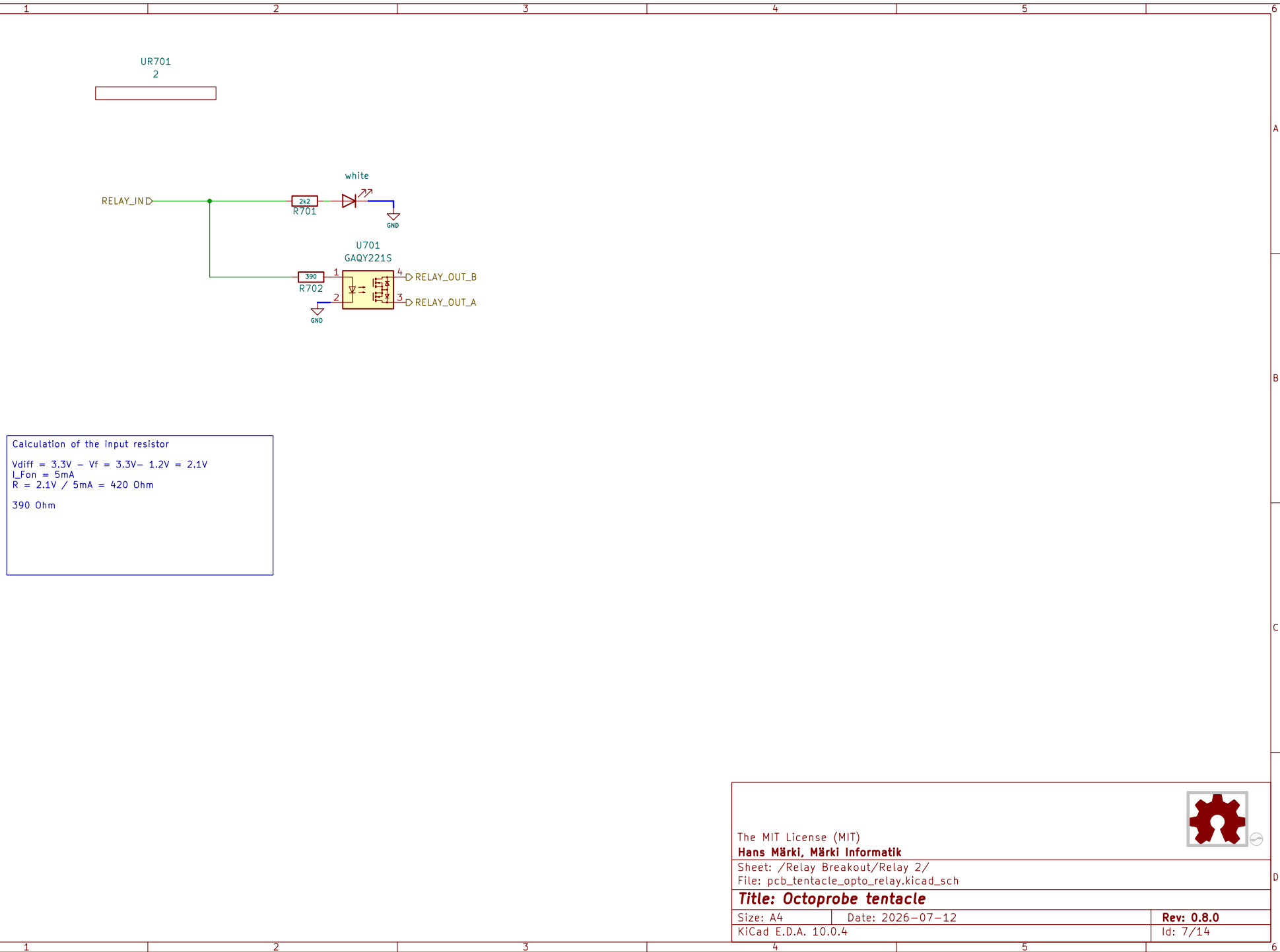
Date: 2026-07-12

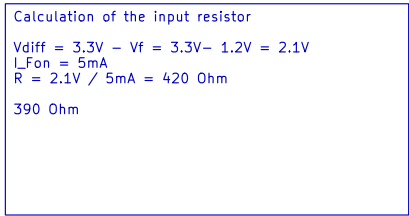
Rev: 0.8.0

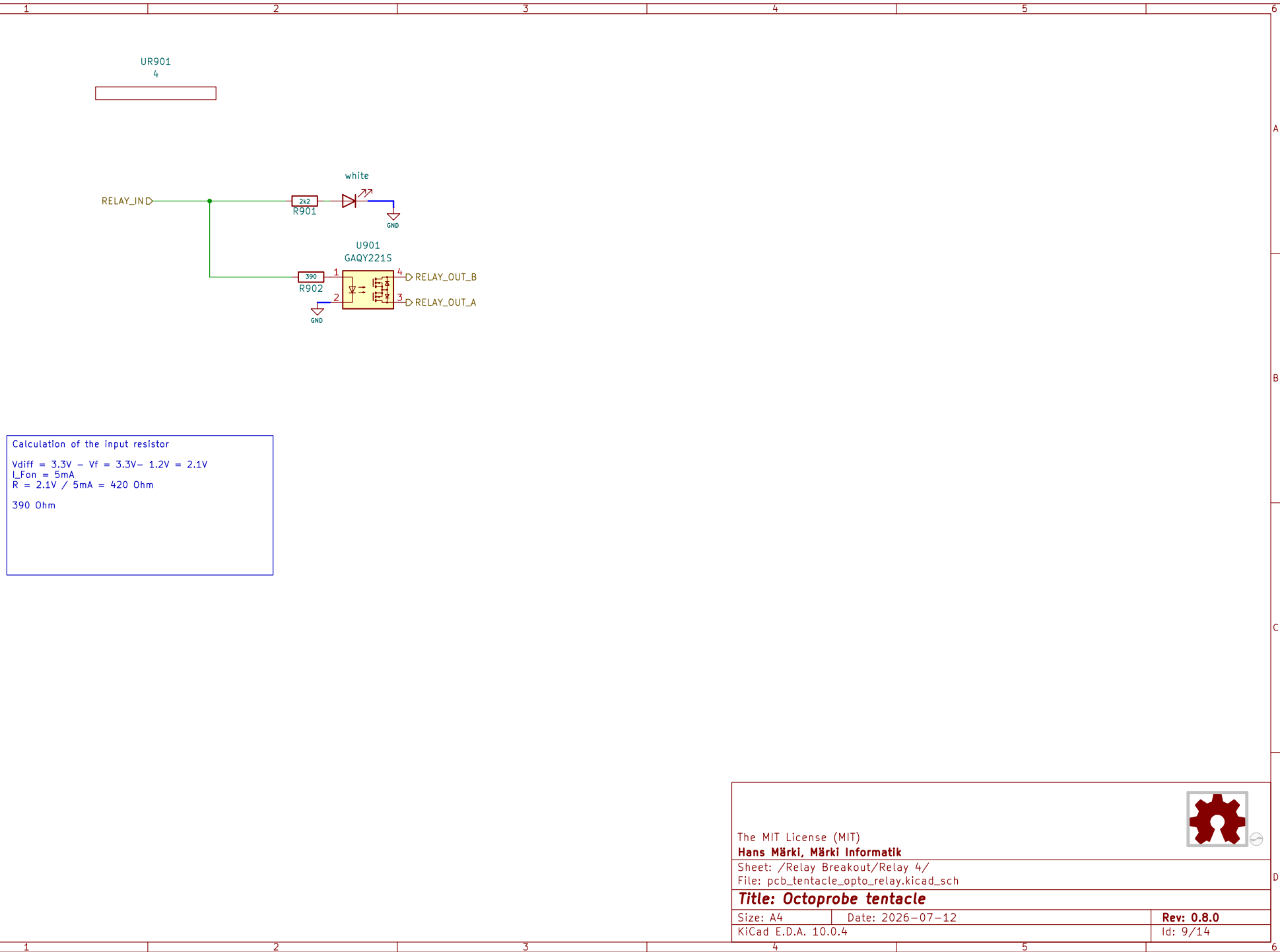
KiCad E.D.A. 10.0.4

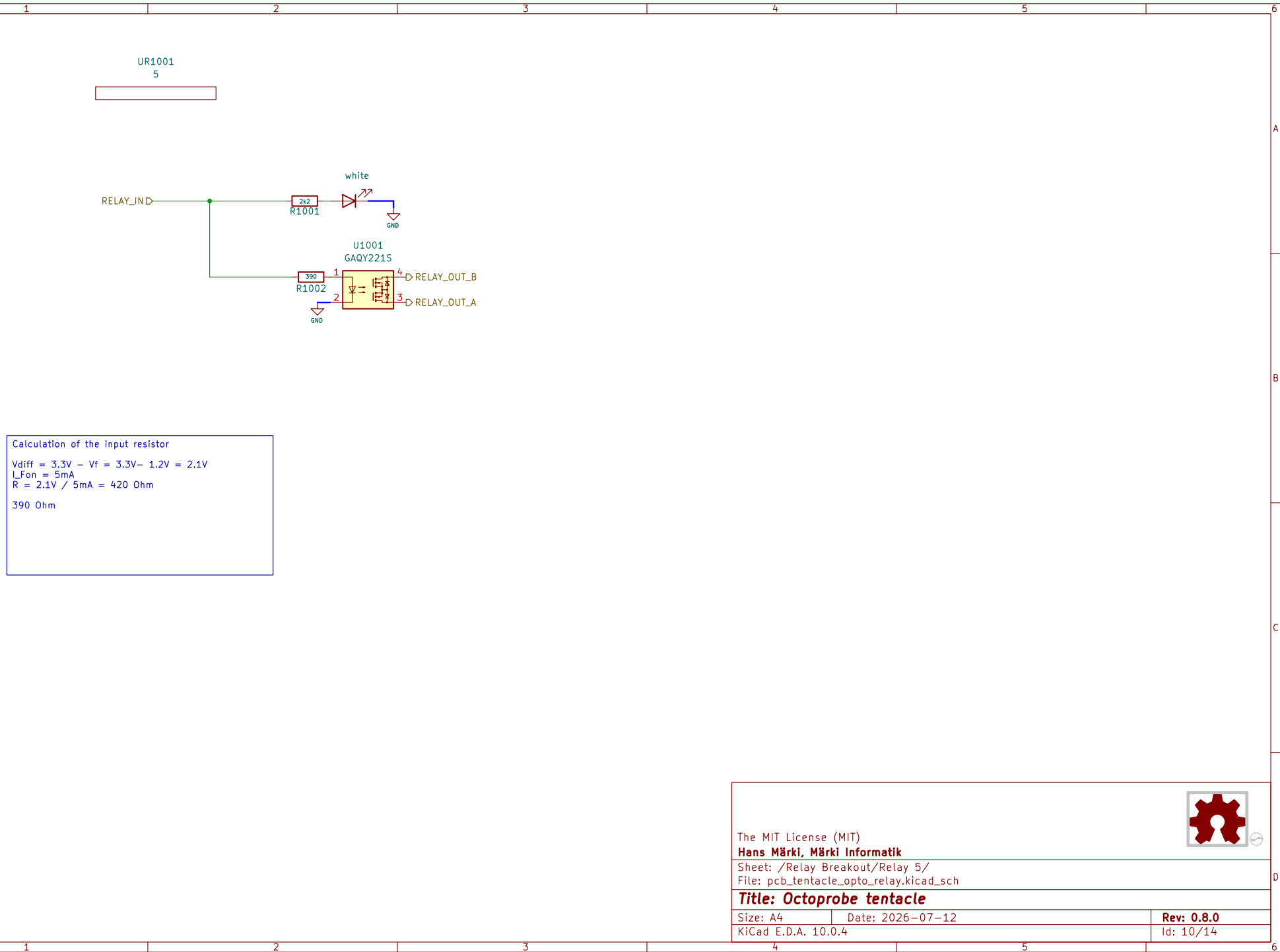
Id: 5/14

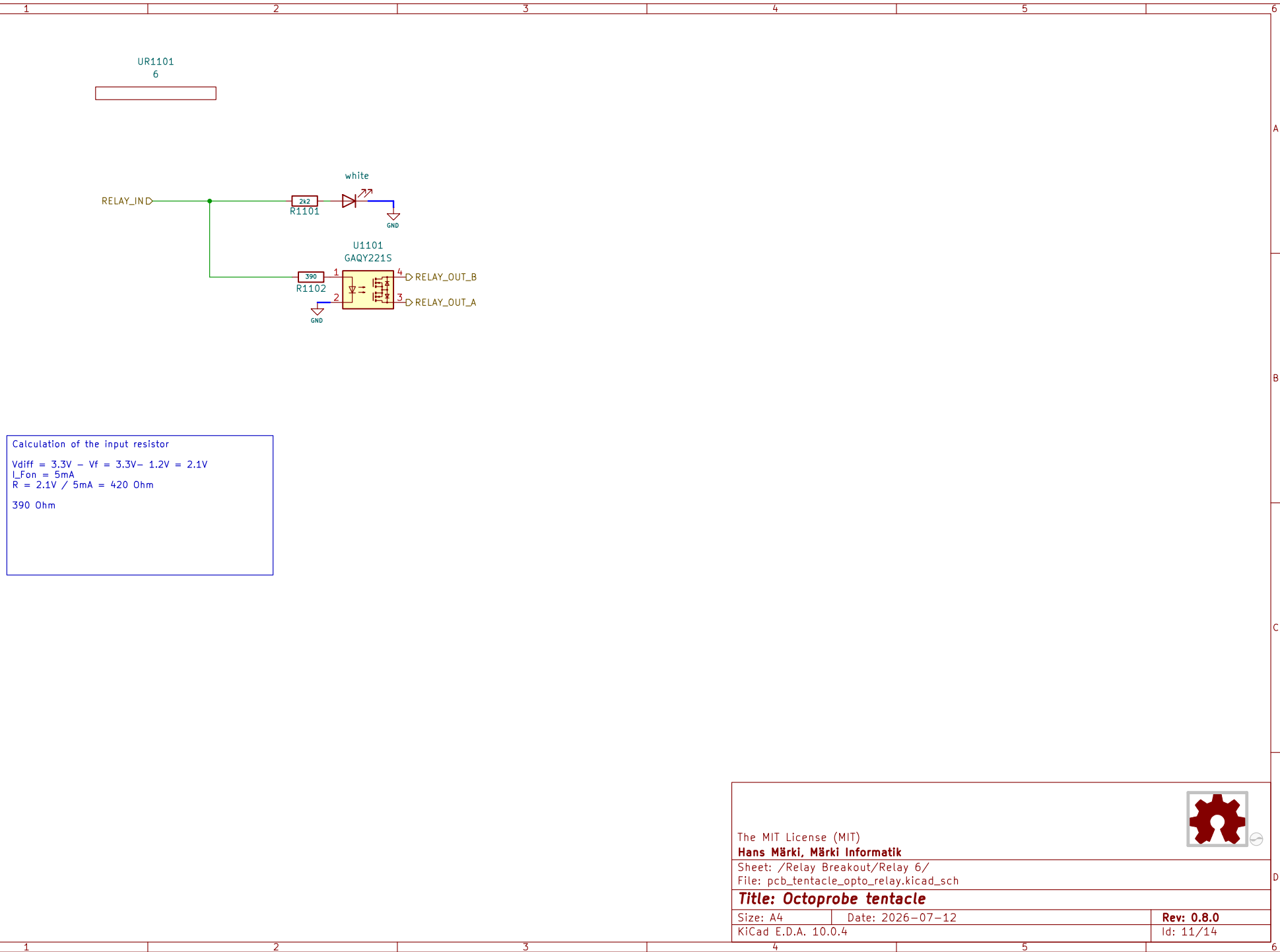


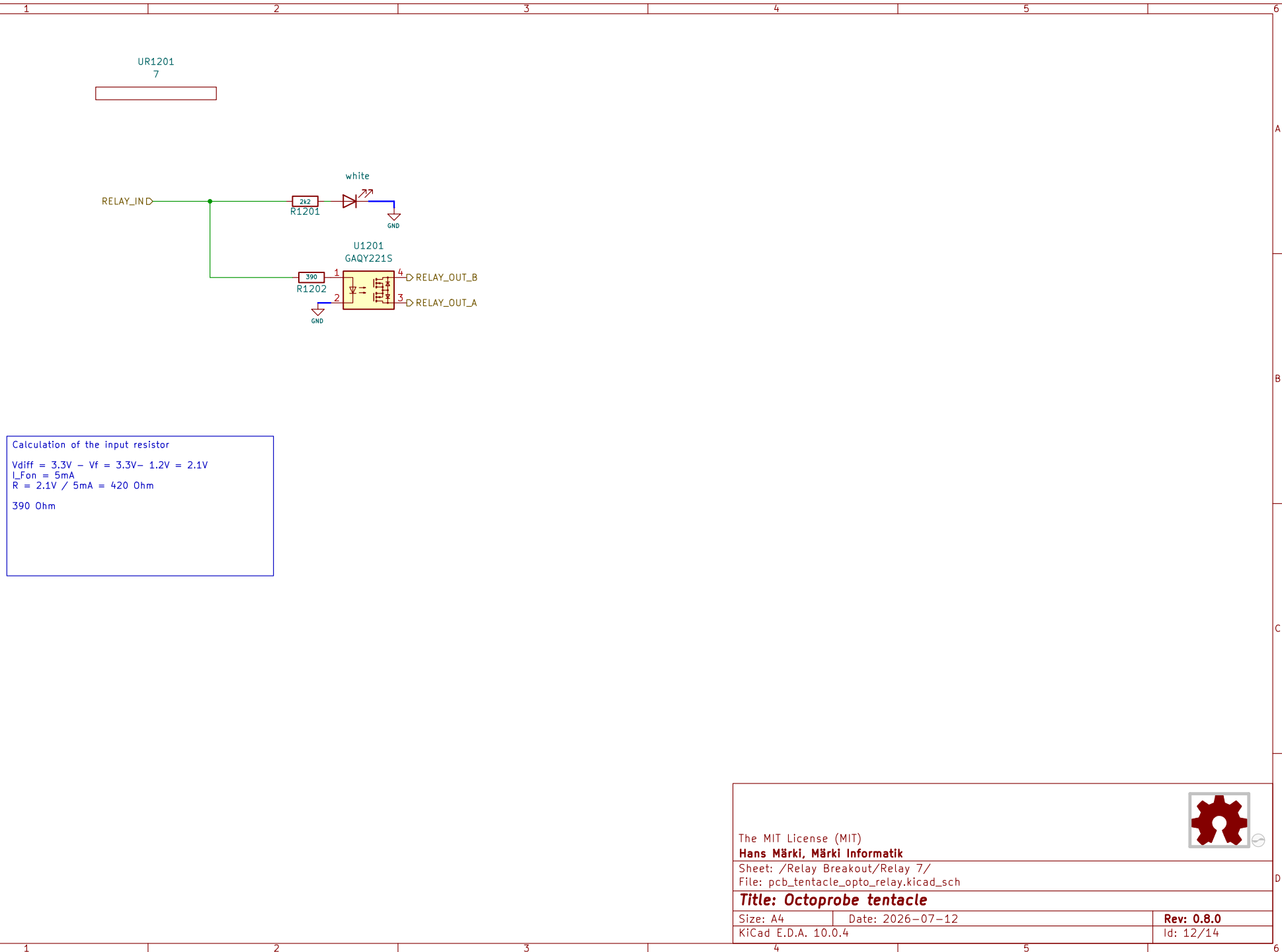






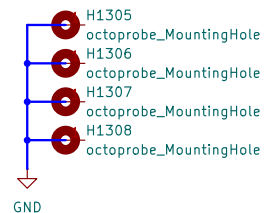




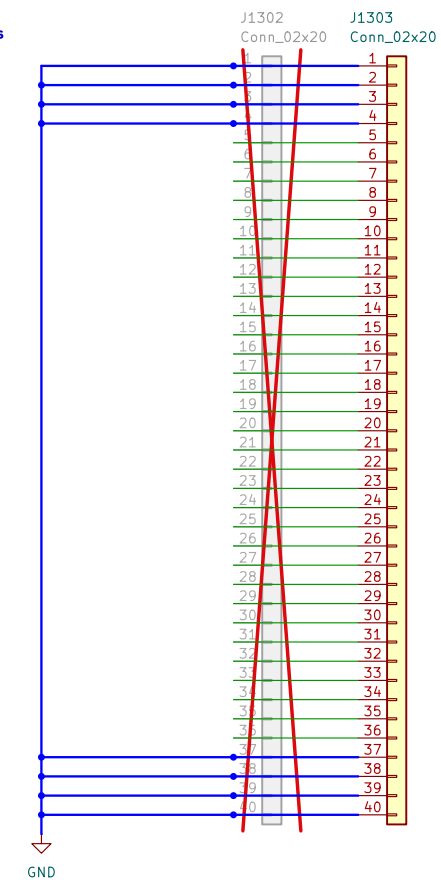


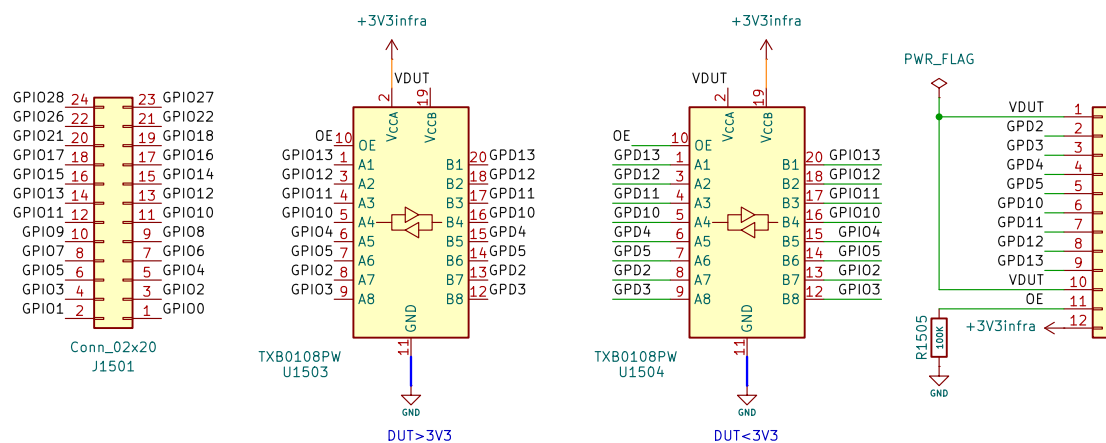
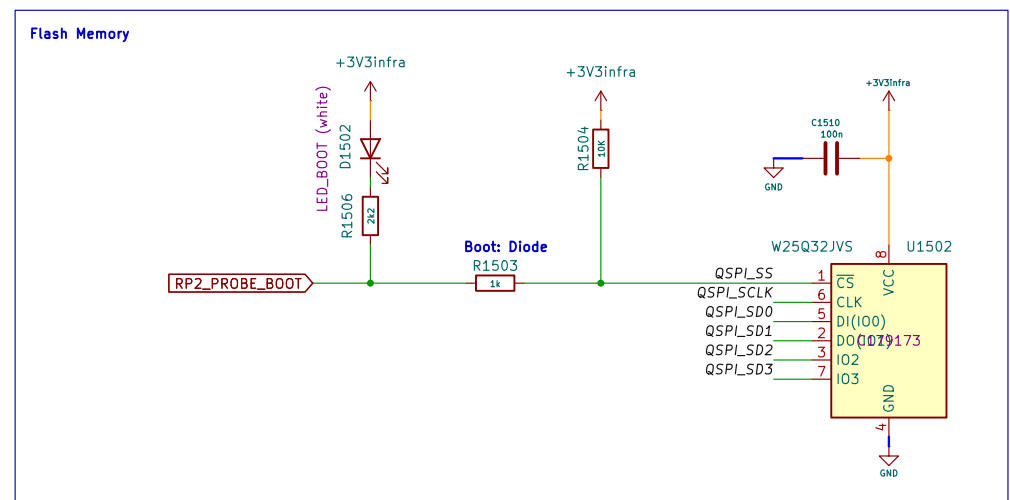
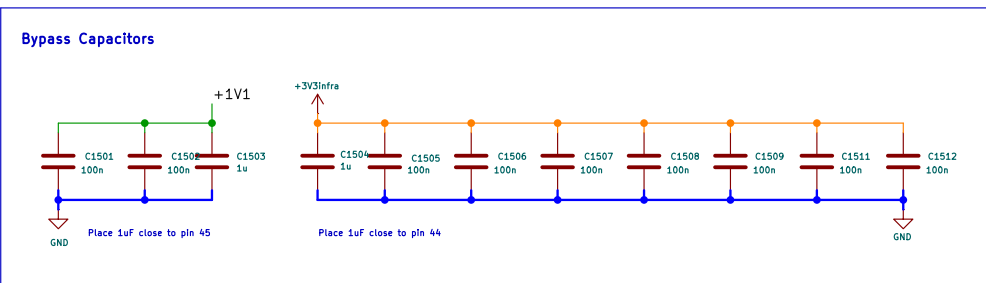
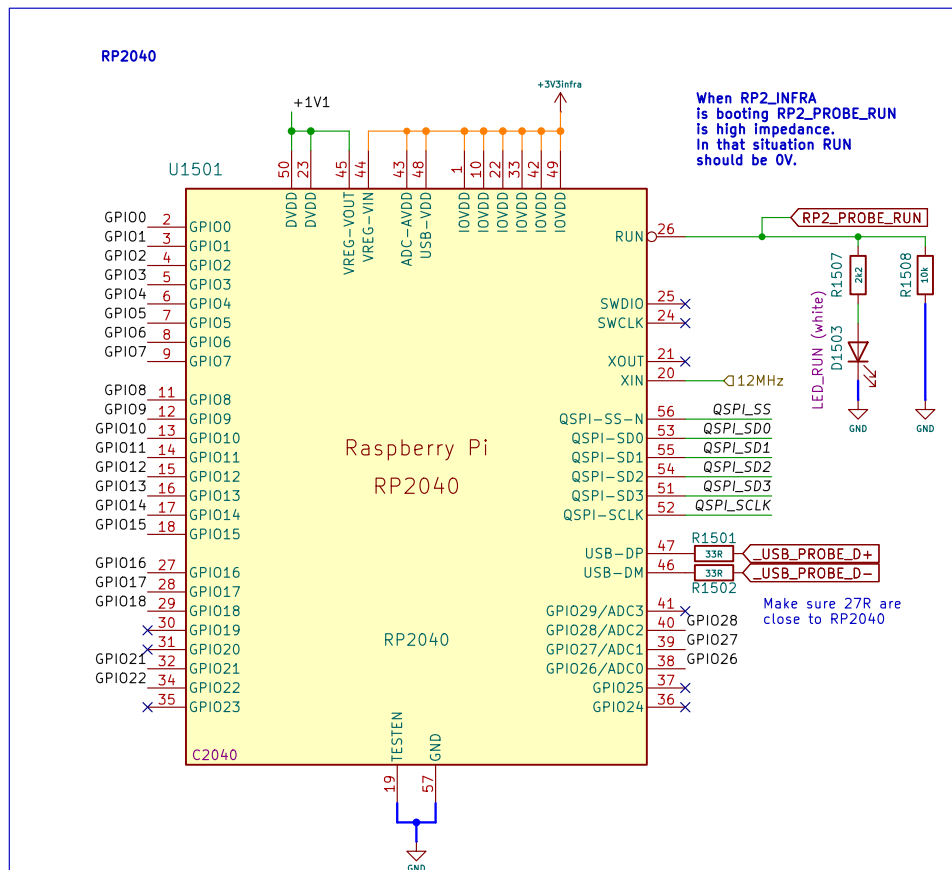
Prototype Area

- TH1301
JLPCB Tooling Hole
- TH1302
JLPCB Tooling Hole

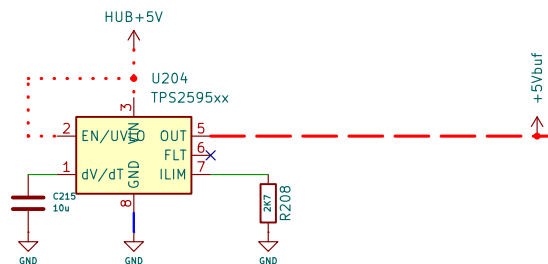


Octopus





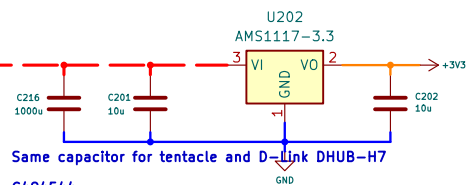
Current Limiter 5V



Datasheet "Active Current Limiting":
Goal: 800mA
 $R_{lim} = 2000 / (0.8A - 0.04) = R2K630$, choosen: R2K7

Datasheet "Slew Rate and Inrush Current Control":
Goal: 5V/1000ms
 $C = 42000 / V * ms = 6'363'636pF = 6uF$, choosen: 10uF

Regulator 3V3

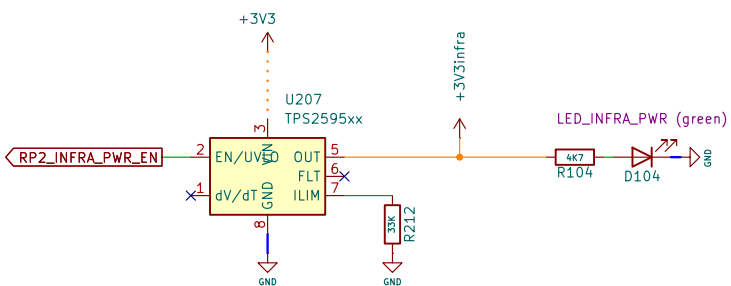


Same capacitor for tentacle and D-Link DHUB-H7

C494544

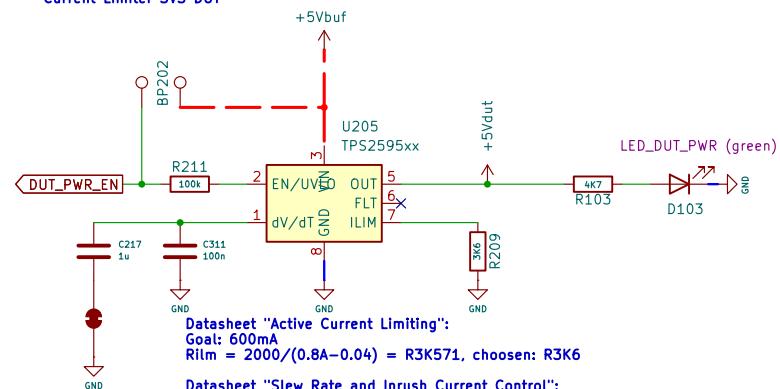
2.5mm distance between wires
6.3mm diameter
11mm height

POWER RP Infra



Datasheet "Active Current Limiting":
Goal: 100mA
 $R_{lim} = 2000 / (0.1A - 0.04) = R33K333$, choosen: R33K

Current Limiter 3V3 DUT



Datasheet "Active Current Limiting":
Goal: 600mA
 $R_{lim} = 2000 / (0.8A - 0.04) = R3K571$, choosen: R3K6

Datasheet "Slew Rate and Inrush Current Control":
Goal: 5V/150ms: 1uF
 $C = 42'000 / V * ms = 1'272'272pF$, choosen: 1uF
Goal: 5/15ms: 100nF

Sheet: /Regulator 3V3/
File: pcb_tentacle_regulator3V3.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 16/14